

ER Diagram

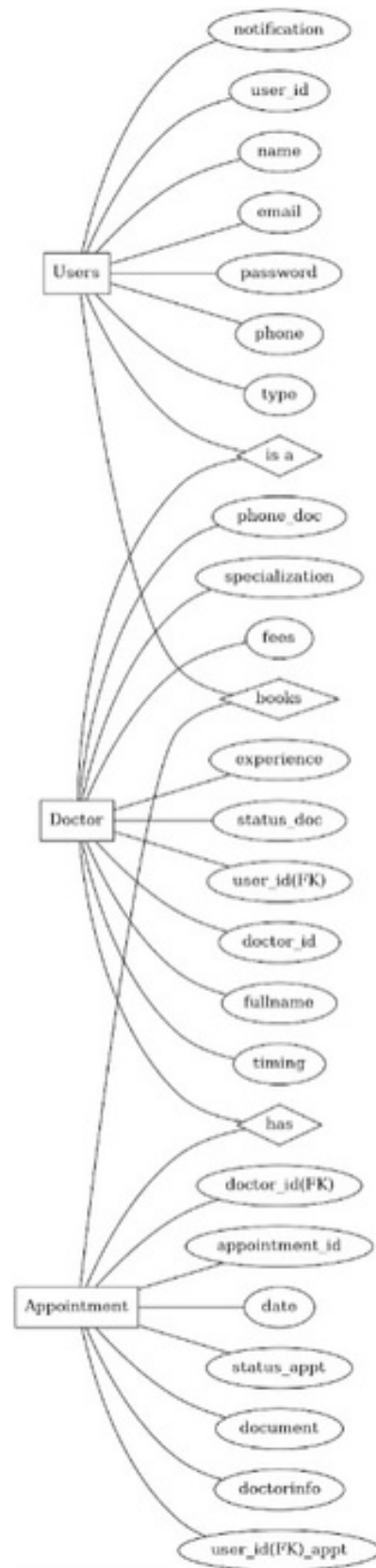


Figure 1: Entity-Relationship (ER) Diagram for Book a Doctor App

Entity-Relationship (ER) Specification

The Entity-Relationship (ER) diagram for the Book a Doctor app represents three key entities: Users, Doctors, and Appointments, with their respective attributes and relationships.

The Users collection holds basic user information, including `_id`, name, email, notification, password, `isdoctor` (to differentiate between patients and doctors), type, and phone. The `isdoctor` field identifies users who are doctors, while others are treated as patients or admins.

The Doctors collection stores information specific to doctors, such as their `_id`, `userID` (acting as a foreign key referencing the Users collection), `fullname`, email, timings, phone, address, specialisation, status, experience, and fees. The `userID` links each doctor to their corresponding user account.

The Appointments collection stores details about appointments, including the `_id`, `doctorInfo` (foreign key referencing the Doctors collection), date, `userInfo` (foreign key referencing the Users collection), document (medical records or other files), and status (e.g., pending, confirmed). This collection maintains the relationship between users and doctors for each appointment.

The relationships are as follows: one User can be linked to one Doctor (one-to-one), a User can have multiple Appointments (one-to-many), and a Doctor can handle multiple Appointments (one-to-many). The foreign keys `userID` in the Doctors collection and `doctorInfo` and `userInfo` in the Appointments collection establish these connections, enabling the app to manage the interactions between patients and doctors effectively.